pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

Market Data Tools

WEEK 2 · DAY 1

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Today: market data tools

Every question names a date

2,940 days of the curve, from the repo

14 tenors, the Treasury's labels

5 tools, curve and spreads

Today

1. market_tools over two files
2. A date the file lacks
3. Changes in bp and slopes, checked
4. Four questions over the curve
5. FRED as a tool, live only
6. The licensed series, refused
7. Synthetic, named synthetic

No tool reads the clock. A missing date is an error result. Recorded runs read files only.

The rules for agents

The curve is described, never forecast

Allowed names

the 10 Yr par yield in the file

the change in bp between two dates

the slope in bp

Never called

a forecast or an outlook

a signal

a view on rates

What an agent produces: text, every number from a tool

Allowed names

the agent returned

the saved run

the brief states

Never called

the agent's opinion or view

what the agent believes

the agent's advice

How market_tools works

The problem	An agent cannot be handed a file of 2,940 rows
The closure	Five functions that close over the two frames, each answering one question from the file
The call	The model sends a tenor and a date; the function returns one number with its unit in the key
You control	The files, the tools offered, and the units in the keys
What comes out	A dictionary of five tools to offer beside the others

market_tools: five tools over two files

market_tools(curve, spreads)

`treasury_yield`	one tenor, one date: yield_pct
`curve_on`	every tenor, one date: yield_pct rows
`curve_change_bp`	one tenor, two dates: change_bp
`slope_bp`	long less short, one date: slope_bp
`synthetic_spread`	IG or HY, one date: spread_bp

Why files

A tool result goes into the next request

The request's key is its body

A live source changes the result,

so the key, so the replay stops

A committed file returns the same

number on any machine, any year

The frames never leave the closure: the model sees only what a tool returns.

Every setting of the five tools

Tool	Setting	Type	Choices	Default
`treasury_yield`	`tenor`	string	-	required
`treasury_yield`	`as_of`	string	-	required
`curve_on`	`as_of`	string	-	required
`curve_change_bp`	`tenor`	string	-	required
`curve_change_bp`	`start`	string	-	required
`curve_change_bp`	`end`	string	-	required
`slope_bp`	`short`	string	-	required
`slope_bp`	`long`	string	-	required
`slope_bp`	`as_of`	string	-	required
`synthetic_spread`	`kind`	string	ig, hy	required
`synthetic_spread`	`as_of`	string	-	required

11 settings, none with a default. The model writes every argument. Tune first: the date.

One bad setting for each tool

Tool	Arguments	What comes back
`treasury_yield`	`10Y, 2026-10-02`	no tenor '10Y'; the file's tenors are ['1 Mo', '1.5 Month',...
`treasury_yield`	`10 Yr, 10/02/2026`	no row for that date in the curve file: 10/02/2026 10 Yr
`curve_on`	`2026-10-03`	no row for that date in the curve file: 2026-10-03
`curve_change_bp`	`10 Yr, 2026-09-02, 2026-10-03`	no row for that date in the curve file: 2026-10-03 10 Yr
`slope_bp`	`2 Yr, 40 Yr, 2026-10-02`	no tenor '40 Yr'; the file's tenors are ['1 Mo', '1.5 Month...
`synthetic_spread`	`xx, 2026-10-02`	kind must be ig or hy

A date written 10/02/2026 gets the same words as a Saturday: no row. That is why the description gives the format.

How a dated lookup works

The problem	A dated question has one right answer or none
The lookup	Find the day among the file's rows; return its value, or an error result
Not asof	pandas asof takes the last row on or before the date: right for a dashboard, wrong for one day
You control	What a missing row returns: here an error that names the date
What comes out	A number, or an error the model reads like any other result

A date the file lacks

```
{'error': 'no row for that date in the curve file: 2026-10-03 10 Yr'}
asof 2026-10-03: 5.28 percent, the row of 2026-10-02
```

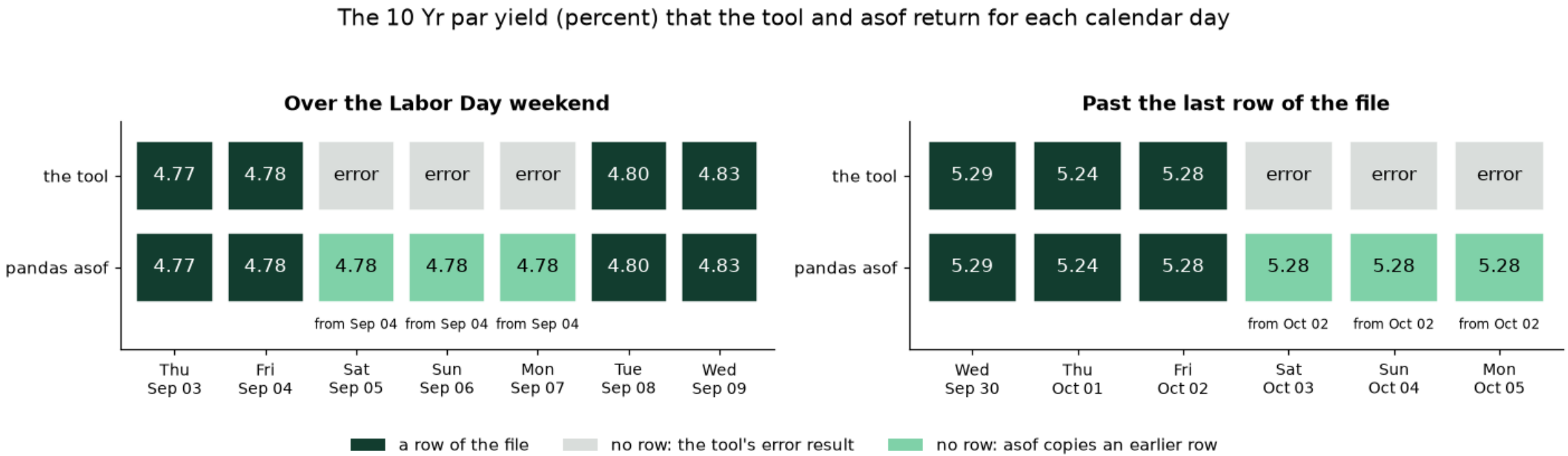
The agent, in the saved run: treasury_yield, then the answer

"The 10 Yr Treasury par yield on 2026-10-03 is not available in the curve file."

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 2,064 prompt and 74 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

The tool answers for the date asked, or says it cannot. Never the nearest row.

The tool and asof, day by day



Six days have no row. The tool says so six times; asof returns a yield six times, the last three for days after the file ends.

How a change and a slope work

The problem	A change compares two dates on one tenor; a slope compares two tenors on one date
The arithmetic	A difference of two yields in percent, times 100, in basis points
The reuse	curve_change_bp reads rows through treasury_yield; slope_bp hands a row to Course 3
You control	The tenor, the two dates, and which tenor is long: long less short
What comes out	One number under a key that ends in _bp, rounded to 6 decimals

Changes in bp and slopes, checked

```
{'tenor': '10 Yr', 'start': '2026-09-02', 'end': '2026-10-02', 'start_yield_pct': 4.79, 'end_yiel...
pandas: 49.00000000000002 bp; marketdata.change_since_bp: 49.00000000000002 bp
{'short': '2 Yr', 'long': '10 Yr', 'as_of': '2026-10-02', 'slope_bp': 45.0}
marketdata.slope_bp: 45.000000000000014 bp
```

x 100

a difference of two yields in percent, in bp

same

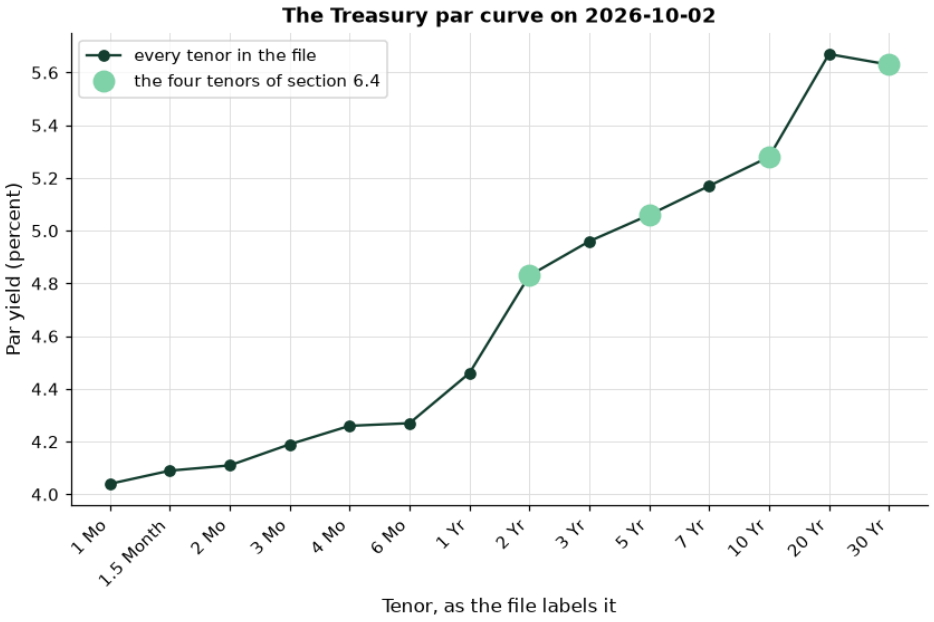
pandas, change_since_bp, slope_bp: one arithmetic

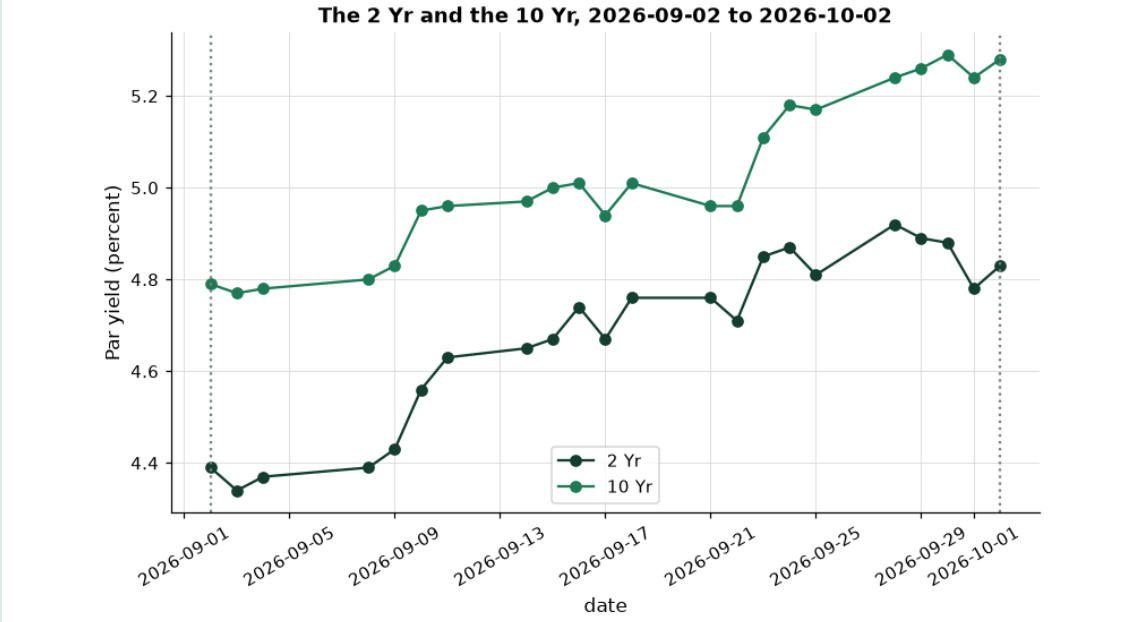
6 decimals

every tool result: the float's tail rounded away

The tool gets checked before any agent uses it.

The curve, the change, and the slope





Right: the rise of the 10 Yr is the change; the gap on the last day is the slope.

Both charts describe the file. They carry no view on rates.

Four questions over the curve

In the saved runs: tools requested, then both checks

```
level   treasury_yield      2 requests  figures 1  passed 1  key 1 of 1
change  curve_change_bp     2 requests  figures 1  passed 1  key 1 of 1
slope   slope_bp            2 requests  figures 1  passed 1  key 1 of 1
curve   treasury_yield x 4  5 requests  figures 4  passed 4  key 4 of 4
```

The slope run's first reply, written beside its tool request, before the tool returned

```
"The 2s10s slope of the Treasury curve on 2026-10-02, which is the 10 Yr par yield less the 2 Yr par yield, is 44 ba..."
```

numbers untraced in it: ['44']; the final answer cites the tool's result

check_figures: the number came from the result it cites. The key: the right question was asked.

How the FRED tool works

The problem	A FRED value can be revised, and the network can be down
The order	Compare the ID with the licensed prefixes; then Course 3's cache; then the date
The cache	Downloaded once, then read from a file in cache/fred/ that Git ignores
You control	Your own key, the cache folder, and the series
What comes out	A value, or an error result

FRED as a tool

fred_latest(series_id, as_of), in this order

1. A licensed prefix: an error result, nothing sent
2. cached_fred_series: once a day, in cache/fred/
3. The date asked, or an error result

Live cells only, with your FRED key

Never in a recorded run

The build had no key, no network

Its saved output:

```
Live cell skipped: no FRED key was found....
```

This product uses the FRED® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.

A FRED value can be revised; the committed file cannot. FRED is for your own live cells.

Every setting of the FRED tool

Function	Setting	Chosen by	What it is
`fred_tool`	`api_key`	you, once	the student's FRED key (from get_secret, never printed)
`fred_tool`	`cache_dir`	you, once	the work folder's FRED cache
`fred_latest`	`series_id`	the model, each request	the FRED series ID, for example DGS10.
`fred_latest`	`as_of`	the model, each request	the date as YYYY-MM-DD, for example 2026-10-02.

Two settings are yours, fixed before any request; two are the model's. No setting turns the licensed refusal off.

How the refusal works

The problem	A refusal only works if it comes before the side effect
The check	The first line compares the ID, in upper case, with the licensed prefixes
The refusal	An error result, and the function ends: nothing downloaded, written, or sent
You control	The prefix list, and where the check sits: first
What comes out	An error text the model can read, and nothing else

The licensed series

```
{'error': "BAMLH0A0HYM2 is licensed data (ICE BofA or Moody's): this tool never downloads it"}
{'error': "BAA10Y is licensed data (ICE BofA or Moody's): this tool never downloads it"}
files in the FRED cache folder: []
```

0

values returned

0

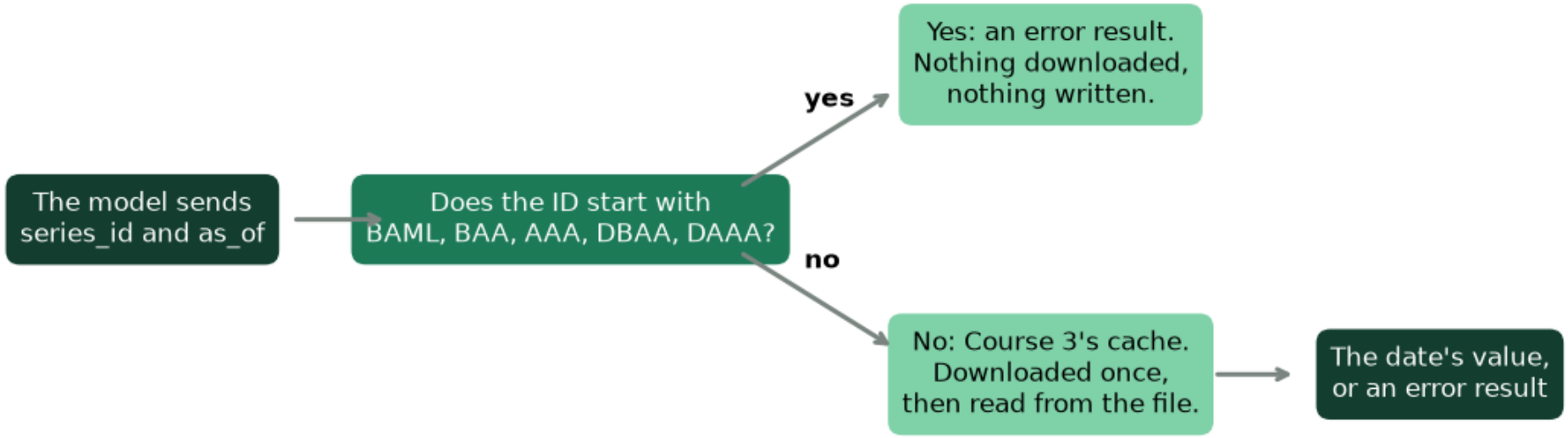
files in the FRED cache

None

the key the tool was given: a refusal sends nothing

Shown by asking the tool directly. A recorded run never holds a licensed series ID.

The refusal comes first



The two licensed IDs never reached the second box: no key, no network, no file.

The synthetic spreads, named synthetic

In the saved run: synthetic_spread, then synthetic_spread

```
On 2026-10-02, the course's synthetic investment grade (IG) credit spread was 97.0 basis points,

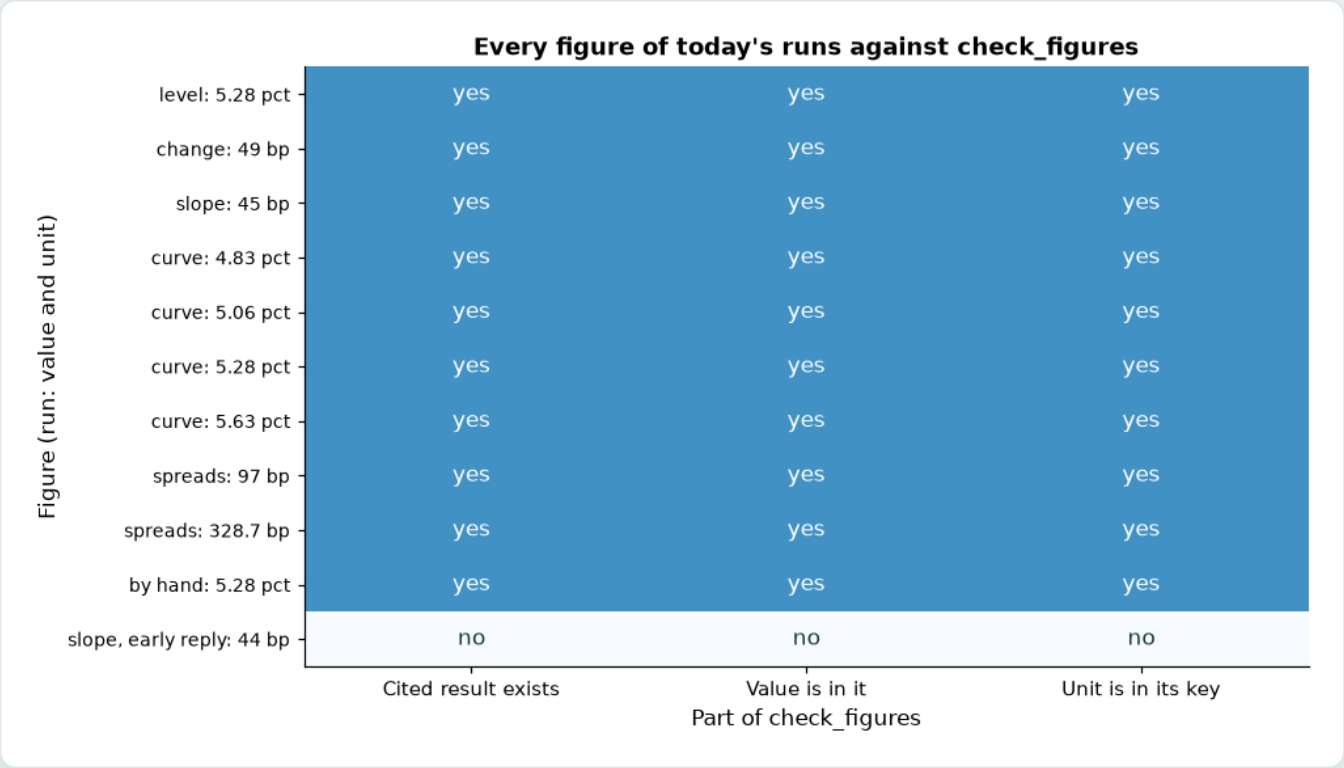
... and the synthetic high yield (HY) credit spread was 328.7 basis points.
```

the answer says synthetic: yes

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 3,257 prompt and 212 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

Invented by the course: no index, no issuer, no real spread series. The word travels with the number.

Every figure against check_figures



Read the grid

Dark: the part passes

Light: it does not

Ten rows dark all the way across

One row light: the slope run's

early reply, written beside the

tool request, before the tool ran

A check on figures catches a number written before the tool that computes it returned.

Excel: INDEX and MATCH, and #N/A

```
=INDEX($B$2:$O$2941,MATCH(DATE(2026,10,2),$A$2:$A$2941,0),MATCH("10 Yr",$B$1:$O$1,0))
  returned 5.28
=(INDEX($B$2:$O$2941,MATCH(DATE(2026,10,2),$A$2:$A$2941,0),MATCH("10 Yr",$B$1:$O$1,0))
  -INDEX($B$2:$O$2941,MATCH(DATE(2026,9,2),$A$2:$A$2941,0),MATCH("10 Yr",$B$1:$O$1,0)))*100   returned 49.00000000000002
```

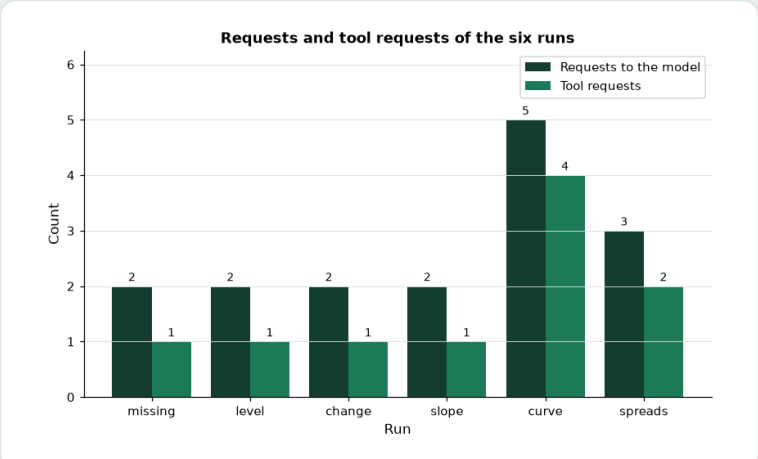
Typed into Excel 16.0 build 20430: 8 of 8 rows the same as the tools

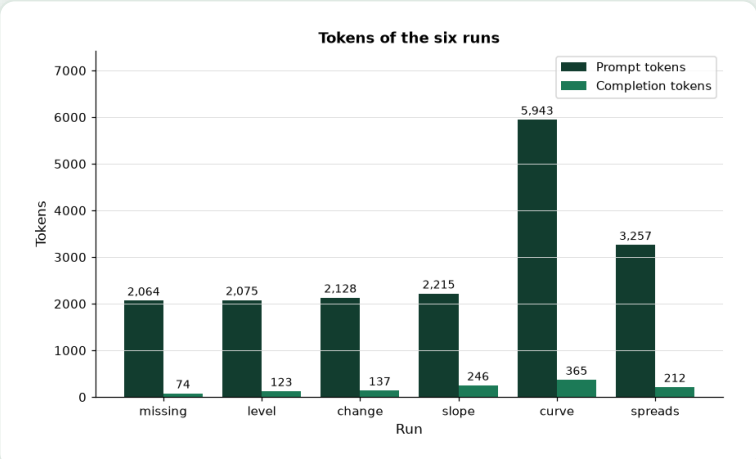
#N/A: 10 Yr on 2026-10-03; 10 Yr change, 2026-09-02 to 2026-10-03; 10 Yr on 2014-06-02

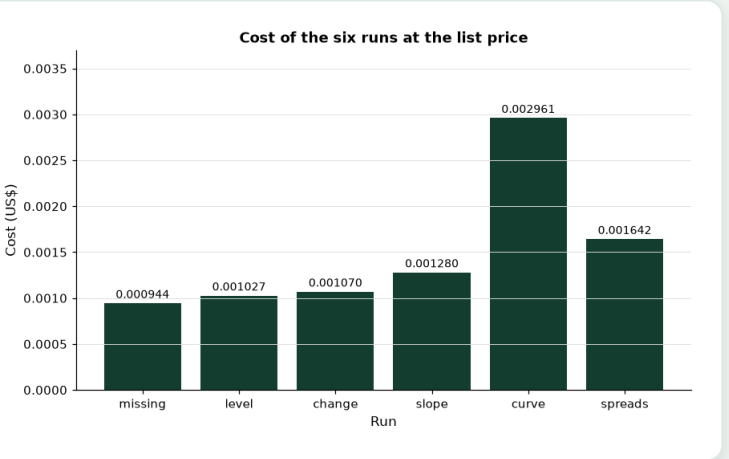
MATCH with 1, or XLOOKUP with -1, on the Saturday: 5.28, the Friday's row

git log --oneline --stat -3: 3 commits, each with its files. 36 passed.

What each run used and cost







Run	Requests	Tool requests	Prompt	Completion	Cost (US$)	Traced	Passed
missing	2	1	2,064	74	0.000944	0	True
level	2	1	2,075	123	0.001027	1	True
change	2	1	2,128	137	0.001070	1	True
slope	2	1	2,215	246	0.001280	1	True
curve	5	4	5,943	365	0.002961	4	True
spreads	3	2	3,257	212	0.001642	2	True

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

Refuse a licensed prefix before anything is downloaded or written.

→

2

Code back

Written for the course in the style of an AI answer. Read the order.

→

3

Test first

A fake download records whether it was reached. Run it: it fails.

→

4

Fix

Refuse first, then Course 3's cached_fred_series.

Today: the refusal comes after the download and the write, so the licensed series lands in the cache.

Code that runs is not the same as code that is right.

What you can do now

- Market data tools over a committed file
- A missing date is an error result
- Changes and slopes checked against pandas
- Figures checked twice: result and key
- FRED live only; licensed refused first

Week 2, Day 2 Document tools: statements, indentures, 8-K sections